STUMPRIDER · MARKET AND VIABILITY

StumpRider market and viability

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Bamboo local variant USD 66.90 after Amish's decision O8 of 2026-10-25
0.1	2026-10-24	Amish Chadha	First issue in the kit 1.9.1 upgrade; benchmark prices checked 2026-10-25

StumpRider market and viability

StumpRider is not a commercial product: it is an open design to be made by village workshops and handed out by child-protection and fisheries partners. Nothing on sale does its job. The nearest bought items are angling lure retrievers (about USD 15 for a 3 oz weight that knocks a lure off a line) and small grapnel anchors (about GBP 13 for 1.5 kg), the working principles of concept options B and C; neither lifts a gillnet off a branch or caps the pull on a canoe. One kit's parts cost USD 95.50 for the aluminium prototype and USD 66.90 for the bamboo local variant, against a target of USD 40 (PR-010, not met). Prices are cited with the date checked; no price here is invented. Method: STANDARDS section 23.

> **Safety:** The tool must never be sold or promoted as a reason to keep children on canoes, or as safety equipment for divers. It goes out only with child-protection work.

Segment and users

- **Who:** canoe crews and boat owners who set gillnets among drowned trees on Lake Volta, Ghana; possibly other reservoirs with standing forest (Lake Kariba to confirm, SMR-PRB-001).
- **Who pays:** child-protection and anti-trafficking NGOs, fisheries programmes and donors, in the first instance; boat owners once the tool has proven itself.
- **Scale:** IJM's 2024 study found 15 % of children working in Lake Volta fishing reported diving to retrieve nets (cited in SMR-PRB-001). The number of canoes that set nets among trees is not known: to assess with the partner.

Competitive benchmark

Table 1. Benchmark.

Item	What it does	Price	Source and date checked	Gap
Knockout Fishing Ott-O-Retriever (lure retriever, about 3 oz)	A weight that slides down a fishing line and knocks a snagged lure free	USD 15.49	Tackle Warehouse, https://www.tacklewarehouse.com/Knockout_Fishing_Ott-O-Lure_Retriever/descpage-OTTO.html, checked 2026-10-25	Sized for a single line and lure; no lift, no pull cap (option B's principle)
1.5 kg galvanised folding grapnel anchor	Small-boat anchor; also used to drag for lost gear	GBP 12.99	GS Products, https://gsproducts.co.uk/1-5kg-galvanised-steel-folding-boat-anchor, checked 2026-10-25	Bites trees as readily as nets; no pull cap (option C's principle)
Cutting the net	Frees the canoe at once	Cost of the lost net section: to assess with partner crews	Not available	Loses the net and catch
Sending someone to dive	Frees the net	No cash cost to the boat owner	SMR-PRB-001	The harm the tool exists to remove

Estimated cost per unit

From the BOM (`bom/bom.csv` , `bom/bom-bamboo-variant.csv` , prototype prices) and SMR-CAL-001 s9 and s10:

- Aluminium prototype kit: USD 95.50 [C1]; the pole, sleeves and rivets are 30 %, galvanising 13 %, the crutch 13 %.
- Bamboo local variant kit, its own crutch and one jigging weight shared by five canoes (decision O8): USD 66.90 [V8].
- Smith's labour (about 8.1 h a kit) is not in the parts cost.
- Concept options (SMR-OPT-001): B about USD 52 before its eye bolt, and C about USD 45 before its shackle and rope, estimates built from the same BOM line prices (the eye bolt, shackle and rope prices are to assess).

Value proposition

For a crew, a net freed in minutes without diving and without cutting; for a partner NGO, a practical tool that removes the reason a boat master gives for sending a child down; for a fisheries office, a safe practice to promote. The value lies in the diving that stops, which cannot be priced and is not offset against the kit cost.

Open, non-commercial route

The drawings, model, BOM and build plan are published under CERN-OHL-S-2.0. The route is: a partner NGO and a landing-site committee commission a village smith to make a batch of kits from the build plan (SMR-BLD-001), the partner hands them out with training and its protection work, and smiths at other landings copy the design from the drawings. No sales channel is planned. Who funds the first batch is to assess with the partner (SMR-PLN-001).